

Student Success Analytics Case Study

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Date: June 26, 2026

Project Overview

This project analyzed a dataset of university students to identify factors associated with student success and academic performance. Microsoft Excel was used for initial data inspection; Python and DuckDB in Google Colab were used for data preparation and SQL analysis; and Tableau was used to develop three interactive dashboards.

The analysis focused on performance via students, programs, courses, and instructors. The analysis also identified students who may benefit from early academic intervention.

Tools Used

- Microsoft Excel
- SQL (Using DuckDB in Google Colab)
- Tableau

Key Findings

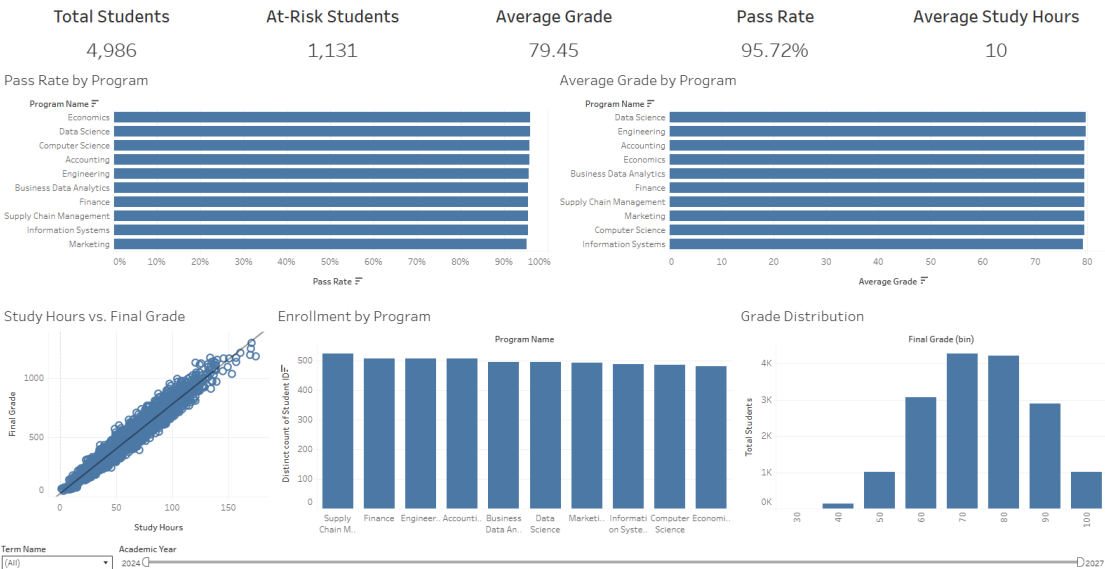
- Analyzed data for 4,986 students.
- Identified 1,131 students at risk of failing (final grade below 60).
- Identified 145 at a high risk of failing (final grade below 50).
- Average final grade of 79.45.
- Overall Pass Rate of 95.72%.

Recommendations

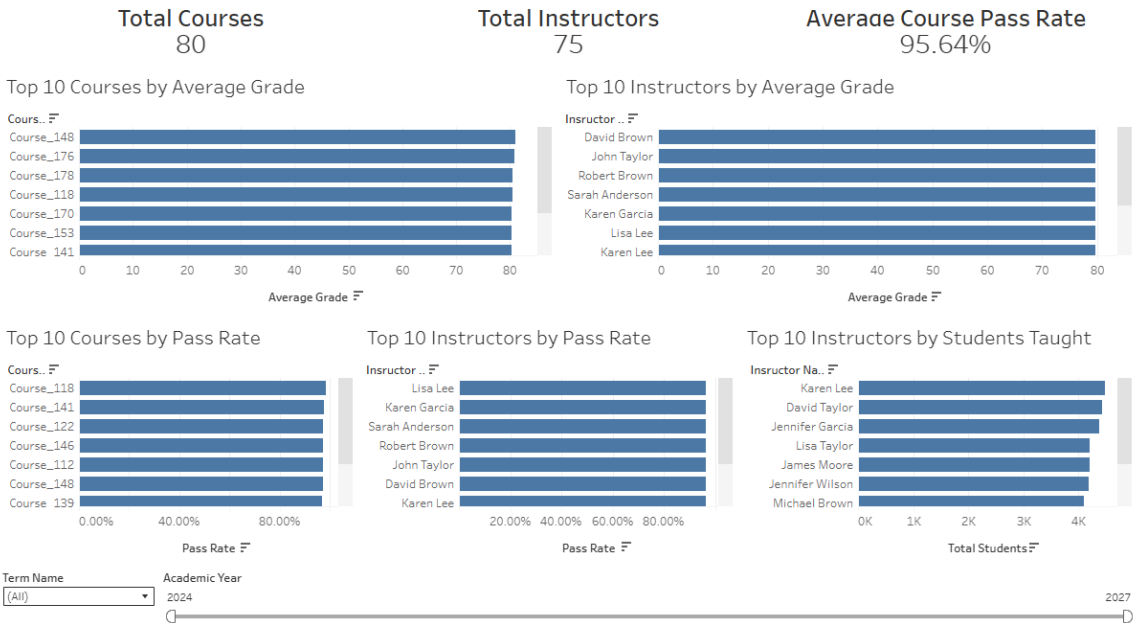
- Expand early intervention programs.
- Offer more ways for students to receive academic support.
- Share best practices among programs.

Dashboards

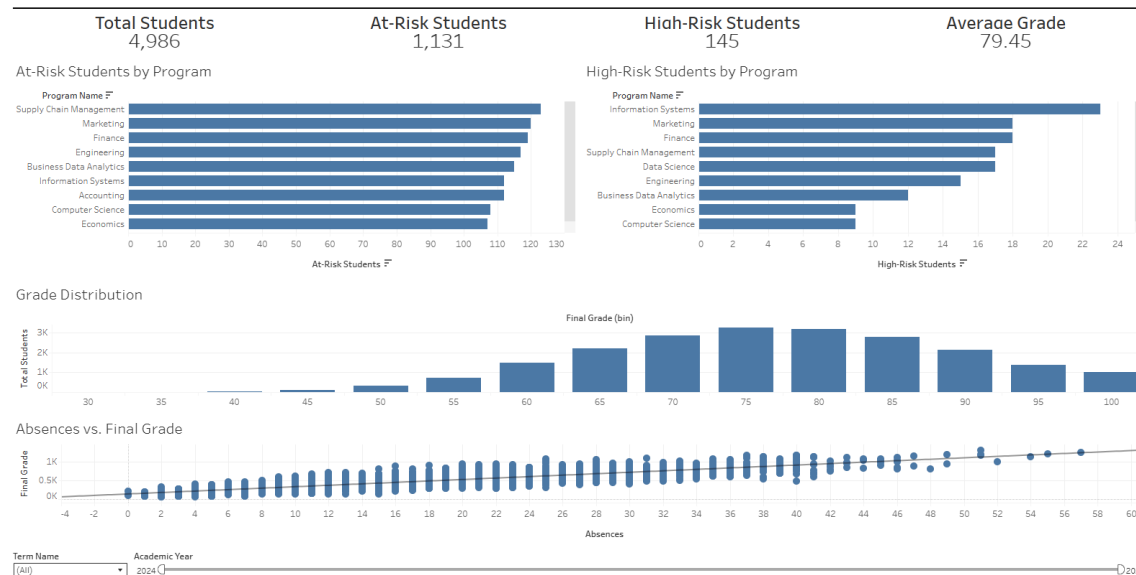
Student Success Overview Dashboard



Course and Instructor Performance Dashboard



Student Risk and Performance Dashboard



Project Outcome

This project demonstrates an end-to-end analytics workflow, including data preparation, SQL analysis, KPI development, and dashboard development. The interactive Tableau dashboards provide stakeholders with an easy way to explore student performance, identify trends, and support data-driven decisions related to student success.

Skills Demonstrated

- Data Cleaning
- Data Modeling (ERD)
- SQL Joins, Aggregations, and CTEs
- KPI Development
- Dashboard Design
- Data Visualization
- Business Analytics
- Tableau
- DuckDB
- Python
- Microsoft Excel

Links

- Interactive Student Success Overview Dashboard
https://public.tableau.com/views/STUDENT_SUCCCESS_DASHBOARD/StudentSuccess?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link
- Interactive Course and Instructor Performance Dashboard
https://public.tableau.com/shared/R8XN4BZZM?:display_count=n&:origin=viz_share_link

- Interactive Student Risk and Performance Dashboard
https://public.tableau.com/views/STUDENT_RISK_AND_PERFORMANCE_DASHBOARD/Dashboard1?:language=en-US&publish=yes&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link
- GitHub Repository
<https://github.com/naguthery/student-success-analytics.git>
- Full Project Report
https://docs.google.com/document/d/16MJRIYAdQ6HB6p92-II3yMn35IWRR1_tAsCfIypZj5k/edit?usp=sharing